BEYOND THE EXPERT: AN ALGORITHMIC APPROACH TO CORRECTING EXPERT BIAS IN FANTASY FOOTBALL PROJECTIONS

Vishnu Datta Jayanti
Senior at Allen High School
Allen, Texas
vishnudattaj@gmail.com

ABSTRACT

While traditional fantasy football projections have long relied on expert intuition, the rise of data-driven forecasting has shifted the analytical landscape. This analysis evaluates four XGBoost models designed for quarterbacks, running backs, wide receivers, and tight ends; the models were trained on a dataset spanning the 2013 through 2023 NFL seasons and validated against the 2024 season. Custom features such as career-maximum performance indicators and lagged inputs were added to the dataset to aid the models in identifying trends and a player's talent ceiling. This research investigates whether "pure" algorithmic models can mitigate the cognitive biases present in the "hybrid" models utilized by industry leaders such as ESPN. Performance was assessed using Mean Absolute Error and Spearman's Rank Correlation Coefficient to benchmark the model against ESPN's preseason projections for the 2025 NFL season. Results indicate that while industry projections maintain superior accuracy in minimizing absolute error, the algorithmic models demonstrate highly competitive ordinal ranking performance. These findings suggest that "pure" machine learning models can serve as a viable supplement to publicly available fantasy football rankings.

***K*eywords** fantasy football · machine learning · sports analytics · bias · feature engineering

Introduction

The transformation of fantasy football into a multi-billion-dollar industry has elevated the importance of preseason projections (Bloomberg, 2026). The methods utilized by experts to form these baselines have evolved dramatically over the years. Traditionally, experts synthesized fantasy football rankings through a blend of box score tracking, trade publications, and direct communication with insiders. However, the rise of the internet fundamentally transformed the work of analysts, expanding the available data from static box-score metrics into massive datasets comprised of advanced statistics (Group, 2024). The complexity of the data surged with the advent of Next Gen Statistics (NGS). A long-term collaboration between the NFL and Amazon Web Services, NGS utilizes RFID chips embedded within the players' shoulder pads and the football itself to capture sub-second positional data (Content, 2019). This flow of "tracking data", in which variables such as top speed, separation, and completion probability were measured, shifted the role of analysts from collecting data manually to building complex algorithms.

The industry standard is currently defined by a "hybrid" framework, which ESPN describes as utilizing "a mixture of statistical calculations and subjective inputs" (Clay, 2025). In this model, algorithmic outputs serve as guidelines that are subsequently "corrected" by human experts to account for any trends the model may have neglected. A key limitation of this system is that it hinges upon humans, who are innately biased, to refine the model's outputs. By injecting subjective judgment into the final projections, the rankings may inadvertently prioritize narrative over facts.

Extensive research has already been conducted on human bias in the context of predictive modeling. Specifically, Helmi Hirsimäki conducted a literature review focused on forecasting inaccuracies within the business industry. Her thesis focuses on instances where human experts attempt to correct model-generated outputs and identifies four primary biases that disrupt forecasting: algorithm aversion—losing confidence in models after a mistake; anchoring bias—over-relying on the first piece of information; optimism bias—producing unrealistically positive estimates; and misinterpretation bias—seeing trends in noisy data (Hirsimäki, 2024). Although her analysis focuses on the business sector, the underlying mechanisms behind these distortions can apply to fantasy football as well.

While Hirsimäki outlines the theoretical cognitive biases that plague human forecasting, Martin Spann and Bernd Skiera provide evidence for this phenomenon in sports: more specifically, soccer. Analyzing data from the German Premier Soccer League, their study evaluates the effectiveness of prediction markets, betting odds, and professional tipsters in maximizing returns in a betting market (Spann & Skiera, 2009). Tipsters are individuals who provide subjective game forecasts based on intuition and experience rather than objective data. Spann and Skiera concluded that while prediction markets and betting odds achieved comparable levels of accuracy, both consistently outperformed the subjective forecasts of tipsters. Although their study benchmarked tipsters' performance against betting odds and predictive markets, their conclusions regarding the limitations of expert intuition can be applied to the methodologies used to calculate preseason projections in fantasy football as well. Their research creates a logical opening to test whether modern algorithmic models can similarly outperform the expert-adjusted projections used by industry leaders like ESPN. This research aims to evaluate whether purely algorithmic models can mitigate the inherent biases of hybrid forecasting systems to provide more reliable preseason projections for fantasy football players.

Materials and Methods

The machine learning algorithm used for this study is eXtreme Gradient Boosting (XGBoost) (Chen & Guestrin, 2016). In comparison to other traditional machine learning models, XGBoost is unique in that it can handle unbalanced datasets (Imani et al., 2025). This is necessary, as the NFL's production distribution is heavily skewed. Typically, the vast majority of players are "average". For example, there are hundreds of wide receivers who record 200-400 yards in a season, but only a handful are capable of recording 1500+ yards. This is problematic for most machine learning algorithms as they become "blind" to high performers, treating them as statistical anomalies. XGBoost mitigates these errors through gradient boosting, a technique where multiple weak decision trees are trained sequentially (Clark & Lee, 2025). In this framework, each subsequent tree is specifically designed to minimize the residual errors of its predecessors, iteratively refining the model's accuracy. This approach differs from Random Forest, a model I initially considered, which utilizes bagging to build independent trees in parallel (IBM, 2021). While bagging reduces variance by averaging results, gradient boosting allows XGBoost to capture complex, non-linear relationships within the dataset more effectively (Lev, 2022). Therefore, the model is well-suited for the highly volatile nature of NFL player performance.

To account for positional differences, four XGBoost regressors were developed, each utilizing a comprehensive dataset featuring over 7100 data points spanning the 2012–2024 seasons (Hyde, 2024). The following hyperparameters were optimized to balance both efficiency and model capacity: *n_estimators* was set to 400 to allow for sufficient iterations while preventing overfitting; a *learning_rate* of 0.05 was implemented to ensure that models don't overcorrect based on previous iterations; and *max_depth* was restricted to 3 to limit model complexity and avert overfitting (XGBoost Parameters — Xgboost 1.5.2 Documentation, 2022). To maximize computational efficiency, the *tree_method* was set to *hist*, while the *multi_strategy* utilized a *multi_output_tree* to capture the underlying correlations between the various outputs being measured. Furthermore, *subsample* and *colsample_bytree* were both maintained at 0.7 to prevent any feature from dominating; *gamma* was set to 1, allowing the tree to partition conservatively; and the *objective* was selected as *reg:squarederror* to minimize the mean squared error. Finally, a fixed *random_state* was used to maintain the reproducibility of this study. Overall, these hyperparameters were chosen with the intent of preventing overfitting and balancing computational efficiency.

In order to maximize the predictive signal extracted from the raw dataset, custom features were implemented. Beyond standard counting statistics (e.g., total yards, touchdowns), several time-series aggregate methods were incorporated as well. For example, a three-year rolling window was implemented by creating a lagged variable for every feature within the dataset. By utilizing the 'Past-n' prefix to denote the specific year of the temporal lag, the features allow the models to evaluate a player based on their trajectory rather than a single data point.

While lagged features capture recent performance, they can fail to account for a player's inherent talent. To address this, 'career_max' features were generated for rushing, receiving, and passing yards. These features prevent the models from becoming overly sensitive to recent variance. While a three-year rolling window captures short-term trajectories, exclusive reliance on these features risks the models rendering a mid-tier player at their peak and an elite player in a slump as statistically identical.

For example, consider a scenario where Wide Receiver A and Wide Receiver B have near-identical production across the Past-1, Past-2, and Past-3 features, such as three consecutive seasons of 800 receiving yards. A model restricted to a three-year window would view these players as identical assets. However, if Player A has a career max of 1,500 yards while Player B's career max is 850 yards, the model is given the context to distinguish Player A as the more talented one out of the two. The career max serves as an indicator of a player's "ceiling," signaling to the XGBoost regressor that Wide Receiver A is much more likely to explode back into high-tier production than Wide Receiver B.

To capture these nuanced distinctions without introducing personal bias, this research expands the feature set beyond a few selected metrics. In an effort to remain objective, I chose to avoid the "hand-picking" of data. Instead of relying on a pre-defined set of features, the model was given access to 1,933 unique variables. By utilizing the inherent feature selection capability built into XGBoost, the model autonomously identified and prioritized attributes with the highest predictive utility. This allows the model to project based on statistical correlation rather than heuristic bias.

Once the feature space was defined, the model was moved into the training and evaluation phase, where the updated dataset was split into a training and testing split. The models were trained on historical data spanning from 2012 through 2023, while the 2024 season was used to evaluate the models on unseen data, where the Coefficient of Determination was monitored to assess the models' fit. Rather than predicting fantasy points as a whole, the models were trained to forecast each component of the fantasy scoring formula independently. Following the inference phase, these individual stat projections (e.g., projected yards, touchdowns, and receptions) were aggregated and used in modified versions of ESPN's Points Per Reception (PPR) formula to generate a final fantasy point total, which was then used to create the models' rankings (Scoring Formats, 2014). This component-based approach ensures that the models capture the specific scoring nuances of the PPR format instead of attempting to predict the fantasy point total directly. Noisy outputs were removed from each formula, such as receiving statistics for quarterbacks (QB) and passing statistics for skill players (i.e., running backs (RB), wide receivers (WR), and tight ends (TE)), ensuring rankings are not skewed by unrelated metrics. The formulas are displayed below.

$$FP_{QB} = \frac{PY}{25} + 4(PTD) + \frac{RY}{10} + 6(RTD) - 2(INT) - 2(FUM) \tag{1}$$

$$FP_{Skill} = \frac{RecY}{10} + 6(RecTD) + REC + \frac{RY}{10} + 6(RTD) - 2(FUM) \tag{2}$$

Where FP_{QB} represents QB fantasy points, FP_{Skill} skill player fantasy points, PY passing yards, PTD passing touchdowns, RY rushing yards, RTD rushing touchdowns, INT interceptions, FUM fumbles, $RecY$ receiving yards, $RecTD$ receiving touchdowns, and REC total receptions.

In order to measure the models' success for the 2025 NFL season, the 2022 to 2024 seasons were extracted as inputs. In this time period, the data from the 2024 season was assigned as 'Past-1' variables, while the 2023 and 2022 seasons were assigned as 'Past-2' and 'Past-3', respectively. The 'career_max' features were simultaneously updated to include the 2024 season. The outputs generated by the models and the preseason consensus from ESPN were recorded as the baseline predictors. Following the conclusion of the 2025 season, the final PPR rankings were scraped to allow for a head-to-head evaluation between the XGBoost regressors and ESPN's pre-season rankings.

In order to measure the performance of the model, three key parameters were taken into account: the Coefficient of Determination (R^2), Mean Absolute Error (MAE), and Spearman's Rank Correlation Coefficient (SRCC). The latter two were used to evaluate ESPN's preseason projections as well, in order to create a competitive baseline. This allows for an objective assessment of the "pure" algorithmic approach compared to the "hybrid" industry standard. R^2 was utilized as a metric to determine how well the model explained the variability in the dataset (Brenndoerfer, 2025). Therefore, R^2 offers a standardized measurement of the effectiveness of the algorithm in transforming raw historical data into a predictive tool. The mathematical formula for R^2 is:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{3}$$

Where y_i represents the actual observed value, $\hat{y}_i$ is the value predicted by the XGBoost model, and $\bar{y}$ is the mean of the observed data.

While R^2 was incorporated as a metric evaluating model features, MAE was used as a baseline to assess the accuracy of the models' predictions. Unlike traditional approaches that measure error in raw fantasy points, this study utilizes MAE as a measure of the deviation in player rankings. By calculating the average distance between a player's predicted rank and their actual end-of-season rank, the metric provides a direct measure of the models' ranking accuracy (Frost, 2025). It is worth noting that MAE is susceptible to outliers. For the MAE analysis, a higher numerical value represents a greater deviation from the actual results, while a lower value indicates superior predictive accuracy. The formula for rank-based MAE is defined as:

$$MAE_{Rank} = \frac{1}{n}\sum_{i=1}^{n}|R_{p,i} - R_{a,i}| \tag{4}$$

Rank-based Mean Absolute Error, where n is the number of players, $R_{p,i}$ is the predicted rank of player i, and $R_{a,i}$ is the actual rank of player i.

On the other hand, SRCC is used to evaluate the "strength and direction of association between two ranked variables" (Laerd Statistics, 2018). In other words, it allows for the evaluation of ordinal accuracy that MAE does not support. While MAE measures the actual distance between the ranks, SRCC measures the extent to which the model maintains the structure of the leaderboard. This is especially important in fantasy football, as the ability to accurately pick the top "tiers" of players is arguably more important for a successful draft than the numerical accuracy of a single ranking. It is worth noting that SRCC is resistant to outliers. SRCC utilizes a scale where negative values represent an inverse relationship and results near zero indicate no correlation; therefore, a higher coefficient signifies a stronger, more accurate alignment between the projected and actual rankings. The mathematical representation of SRCC is expressed as:

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)} \tag{5}$$

Spearman's Rank Correlation Coefficient (ρ), where d_i represents the difference between the predicted and actual rank for each player and n represents the total number of players in the sample.

In using both MAE and SRCC, the evaluation framework attempts to assess all aspects of model performance from a holistic perspective. While rank-based MAE offers a close look at the average "distance" of miss, SRCC attempts to ensure that the models maintain the hierarchical integrity of the draft board. In other words, the models are being assessed not only on their accuracy regarding individual players, but also on their general ability to navigate the tiers of fantasy football. The Python implementation of the XGBoost models, including the training scripts and statistical evaluations can be found in the project repository (Jayanti, 2026).

Results

To evaluate the XGBoost models, I first used R^2 to measure how accurately the regressors captured the variance for each statistical category. The performance across all 9 variables is detailed in Table 1.

Model Fit for Individual Performance Metrics

Statistical Category	QB	RB	WR	TE
Passing Yards	0.42			
Rushing Yards	0.39	0.37	0.23	0.00
Receiving Yards		0.36	0.34	0.37
Passing Touchdowns	0.38			
Rushing Touchdowns	0.42	0.34	0.01	0.01
Receiving Touchdowns		0.08	0.20	0.17
Receptions		0.38	0.33	0.38
Interception	0.20			
Fumble	0.18	0.14	−0.06	0.06

−0.06 0.42

Table 1: Model Fit (R^2) for Individual Performance Metrics

The XGBoost models performed best when predicting yardage across all four positions. These categories consistently produced R^2 values between 0.34 and 0.42, showing a moderate relationship between the features and the outcomes.

Scoring metrics like touchdowns were more difficult to predict and resulted in lower scores near 0.20 or 0.30. Rare events such as rushing statistics for tight ends or fumbles for pass catchers showed very little correlation, often having values near 0.

After analyzing R^2, I examined the features that drove the predictions before benchmarking the final rankings along with ESPN's hybrid projections. Of the 1,933 features employed during model training, the 10 most influential predictors for each regressor are illustrated in the four variable importance plots below.

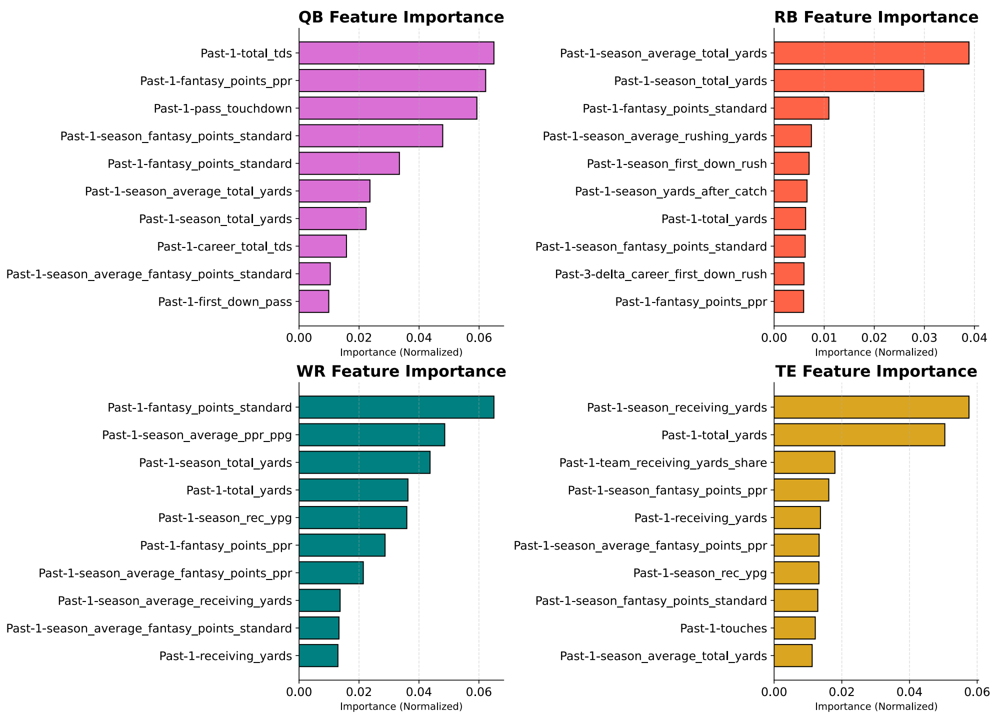

Figure 1: Feature Importance in each model

The feature importance distribution shows that, as expected, there is a focus on recent performance, as evidenced by the dominance of 'Past-1' variables within the top 10. Surprisingly, none of the 'career_max' metrics were present on the graphs, ultimately highlighting the models' preference to incorporate recent data when generating rankings. While the feature importance plots reveal the internal logic of the models, the ultimate measure of their utility lies in their predictive accuracy relative to industry standards. To quantify this, the models' outputs were benchmarked alongside ESPN's preseason projections using MAE; the results are displayed below. To ensure the data remained relevant for fantasy football fans, all metric comparisons were restricted to the top 20 quarterbacks and tight ends, the top 50 running backs, and the top 60 wide receivers.

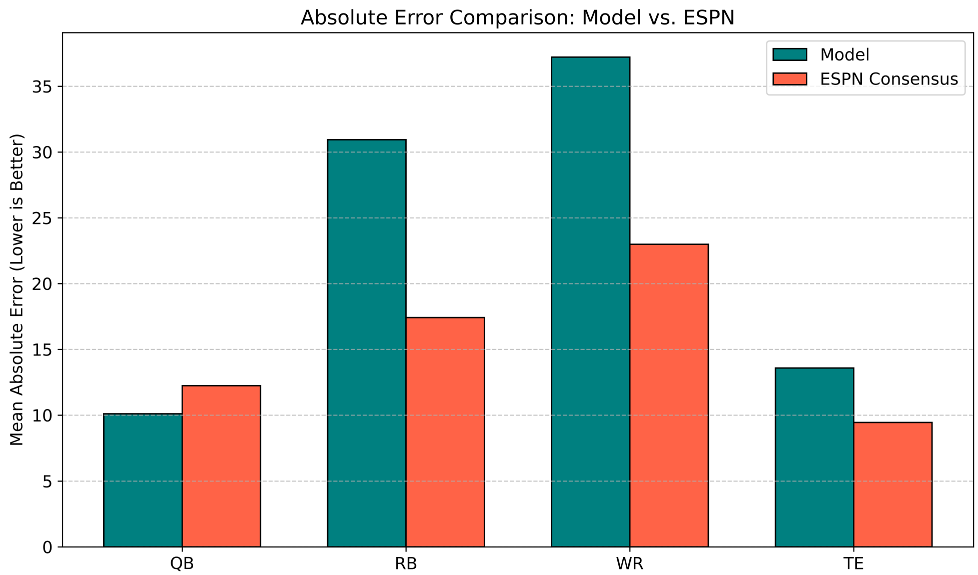

Figure 2: MAE comparison between XGBoost regressors and ESPN

After thorough comparison, it is evident that although the XGBoost regressors had achieved comparable accuracy to ESPN's hybrid projections in the QB and TE positions, a notable gap in performance was present for RBs and WRs. It is worth noting that the algorithmic model had a slight edge over ESPN in the QB category. While Rank-based MAE provides a measure of absolute deviation, it only captures one dimension of model performance. Conversely, SRCC measures how well the model preserved the "tiers" in fantasy football. The degree to which each model preserved these tiers is illustrated in the graph below.

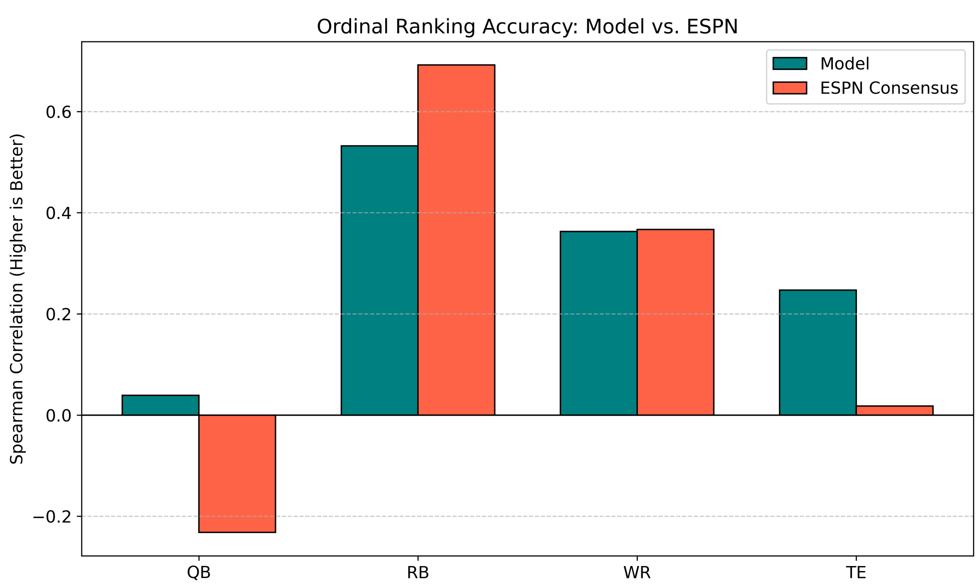

Figure 3: SRCC comparison between XGBoost regressors and ESPN

Once again, the algorithmic model outperformed the industry standard for QBs and TEs. Surprisingly, the model achieved a score almost identical to ESPN for WRs, despite vastly underperforming in MAE. The tight end model demonstrated a significant improvement over ESPN, whose projections displayed a negligible correlation with the final season outcome. A similar trend was observed in the quarterback results; however, it is notable that both ESPN and the "pure" model struggled with this metric, as the XGBoost regressor scored near zero while ESPN's projections had a negative correlation. Conversely, ESPN maintained a substantial advantage in the running back category, mirroring the MAE findings. To ensure the evaluation focused on predictive performance rather than injury luck, six players (who happened to all be RBs and WRs) who failed to play a single game during the season were removed from the rankings. Following their removal, both the MAE and SRCC were recalculated to provide a more accurate reflection of model consistency. The injury-accounted MAE results are illustrated in Figure 4 below.

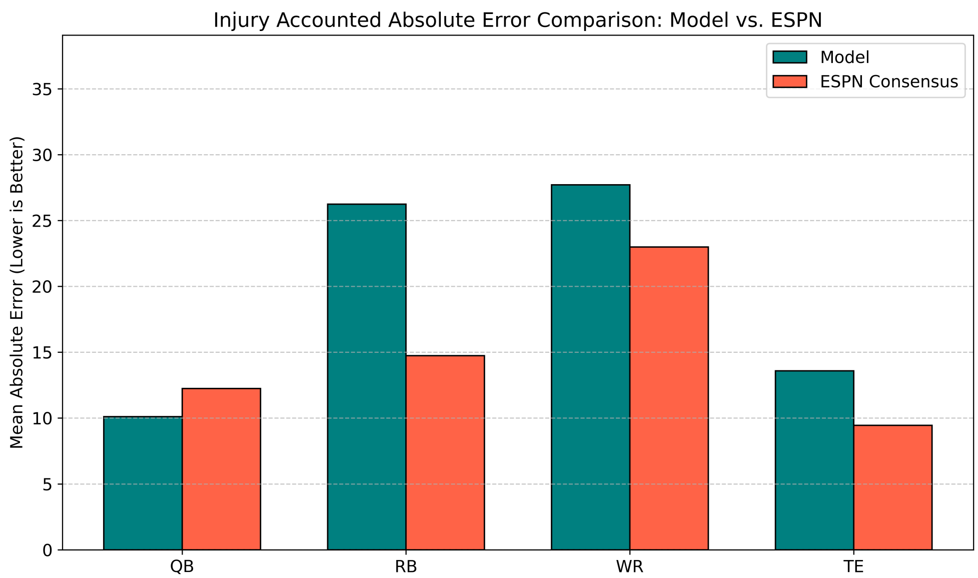

Figure 4: Injury accounted MAE comparison between XGBoost regressors and ESPN

Because the six players removed were exclusively wide receivers and running backs, the results for the quarterback and tight end models remained unchanged from the previous analysis. However, within the running back and wide receiver categories, the gap in MAE narrowed significantly, although both models continued to lag behind the preseason industry projections. Similarly, the recalculated SRCC values, adjusted for player availability, are provided in the following comparison.

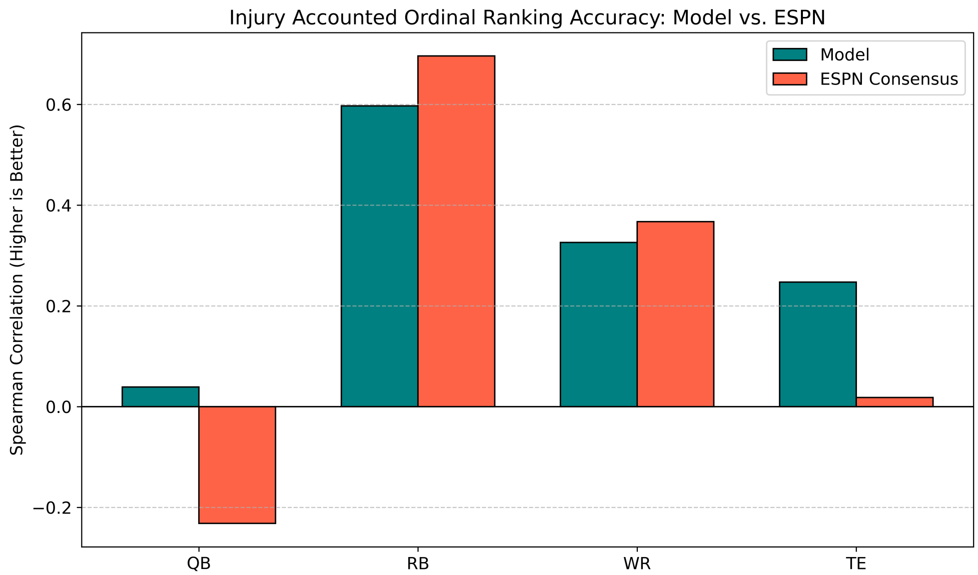

Figure 5: Injury accounted SRCC comparison between XGBoost regressors and ESPN

The recalculated ordinal metrics showed minor improvements for RBs, though they were less dramatic than those found in the MAE analysis. Surprisingly, however, the WR SRCC decreased after accounting for injuries.

Discussion

The results of the XGBoost models provide a clear picture of which player statistics are the most predictable from one season to the next. The R^2 values represent the proportion of variance in each metric that the model was able to explain using historical data. Higher values, like those seen in yardage and receptions, indicate that the model found a strong, repeatable signal in the data. Conversely, values near zero or below suggest that those specific metrics are driven more by random chance than available inputs.

The difference between the values for yards and touchdowns for R^2 demonstrates an important truth regarding football analysis. While yards are a high-volume stat that will always accurately represent a player's contribution and ability, touchdowns are much more subject to small sample size and luck. A player can gain 100 yards through consistent performance, but a touchdown often depends on a single play or a specific coaching decision. This demonstrates why touchdowns are naturally much harder for any model to accurately predict with high precision due to their "noisy" nature and large fluctuations from week to week. The findings of this research also indicate a significant divergence in the effectiveness of "pure" algorithmic models for various positions within the National Football League.

Applying Hirsimäki's framework of misinterpretation bias, it becomes evident why the model outperformed human analysts in QB evaluation. As the most scrutinized position in the NFL, quarterbacks are under constant media pressure, with every throw, decision, and body language cue dissected by fans and pundits alike. This high-stakes environment creates a "noise" of public opinion that often bleeds into professional evaluation. Consequently, variables are frequently used to evaluate QBs that aren't necessarily tied to their actual performance, resulting in a clear form of misinterpretation bias where these external factors are mistaken for the quarterback's individual skill (Bennett, 2025). The model's success over ESPN proves that the "pure" approach was able to mitigate such bias by focusing strictly on objective markers, thereby achieving superior scores over the "hybrid" approach for QBs in both MAE and SRCC. By filtering out the narrative-driven noise that human experts struggle to ignore, the algorithmic approach isolates the true "signal" to create viable QB rankings. However, it is also worth noting that both ESPN and the XGBoost regressor struggled in SRCC, despite the model's superior performance in comparison to ESPN. This highlights the volatility of the QB position in general, as after the "elite" tier of QBs, it becomes increasingly difficult to rank subsequent players who are often subject to high-variance performance (Kartes, 2025).

The next position for which the model achieved notable success was the TE position. Within this category, the model's success was especially noteworthy, as the model dominated ESPN in SRCC and was on par in terms of MAE. Such results may be explained by anchoring bias, a cognitive error in which analysts tend to rely too heavily on the initial information they receive regarding a particular player. Most often, this manifests as an over-reliance on a player's draft capital or physical profile before the draft, as these are often the primary "anchors" for a player's ceiling (Myers, n.d.). Analysts often fall in love with a player's draft stock, be it a blazing 40-yard dash time or a prototypical "big-bodied" physical profile, and often do not adjust their evaluation of a player's ability despite on-field performance indicating a lack of efficiency. The superior rank correlation provided by the model seems to indicate that, by essentially ignoring a player's physical ability and instead focusing on production, it can provide a more accurate reflection of a player's talent.

The cases of Caleb Williams and Kyle Pitts exemplify the conclusions reached above. While ESPN's pre-season evaluations significantly undervalued these athletes, their subsequent performance validated the "pure" models' higher projections. Although the "pure" models also underestimated these players, the margin of error was substantially narrower than ESPN's, demonstrating a more accurate assessment of their elite potential.

Entering the 2025 season, Caleb Williams was highly scrutinized for his rookie season. Picked first overall in the 2024 NFL draft, Williams was a highly sought-after talent who failed to meet expectations. ESPN's QB-specific projections reflected this widespread skepticism, ranking him 17th. In contrast, the model positioned Williams at 10th, which was significantly closer to his eventual 5th-place finish. ESPN's significant undervaluation serves as a clear instance of anchoring bias, where his debut season disproportionately influenced future projections, causing him to plummet in the rankings despite his underlying talent profile.

The evaluation of Kyle Pitts serves to further illustrate these distortions, specifically the misinterpretation bias. After a historic 1,000-yard performance as a rookie, Pitts went on to have three consecutive subpar seasons where he was unable to reach 700 yards. ESPN's rankings responded to this recent performance with a lack of confidence, dropping him to a 17th-place evaluation to start the season. The human evaluators gave in to the emotional response of a multi-year slump, whereas the model understood that a player of Pitts' historical production deserved a statistical chance to perform at a high-end. The model's evaluation of 8th place was vindicated as Pitts went on to have a dominating season, finishing in 2nd place overall. This advantage, however, was not present across all positions.

The two positions ESPN dominated regarding the “pure” model approach were RBs and WRs. In both positions, the model dominated MAE even after adjusting for injuries; similar trends were displayed in SRCC, although the WR model achieved comparable results when using this metric. Running backs and wide receivers are, by nature, difficult to project because their statistical output is not based solely on the player's natural talent, but also on a number of external factors. RBs, for example, are highly dependent upon the strength of the offensive line; in addition, a player's statistical output may vary greatly depending upon whether the offense is running to protect a lead or if they are forced to pass in order to catch up. On a similar note, WR performance is often tied to coaching schemes and the QB. Because the “pure” model wasn't given the context to process these situational variables, it suffered in comparison to ESPN when creating its projections. The "hybrid" method used by ESPN, in this case, may have actually worked in their favor, as the inclusion of qualitative data, such as training camp reports or exclusive interviews, may have picked up on a shift in a player's usage before it was reflected statistically. Outside of predictive accuracy, the "pure" algorithmic approach provides a significant benefit in efficiency. While a sports media giant like ESPN must utilize a large group of analysts to "correct" and improve their projections, which is both time-consuming and expensive, the XGBoost model was able to create comparable rankings at near-zero cost.

These results help identify the limitations of the “pure” model approach, serving as a diagnosis to further iterate upon the model. Chief among these limitations is that the model was only evaluated on two seasons' worth of data, with 2024 being used exclusively to calculate R^2, and 2025 being used to benchmark the model against ESPN. In order to apply the results of the study in a more general context, more testing must be conducted. Another limitation is that the regressor has no understanding of context. In order to truly win at fantasy football, one must do more than identify talent. Many factors influence the performance of a player: these include the players around them, the coaching staff, and their injury history. These are factors that weren't included in the model, representing a gap between purely historical projection and the dynamic reality of professional sports. These are all taken into consideration by ESPN when making their rankings, whether by using more nuanced data or using human experts to reflect these changes. There are, however, ways to mitigate these issues when using the “pure” model approach.

Although quantifying the effect of a new coaching staff or roster change is a complex challenge, it can be addressed by analyzing team statistics. Every year, Mike Clay projects individual player statistics across the entire league before the start of the NFL season (Clay, 2026). While these are presented as player-specific forecasts, they are built on a team-level framework, meaning they account for the context missing in the XGBoost models. By aggregating these individual projections, which means summing the projected passing, rushing, and receiving outputs for all players on a roster, one can calculate a projected total for team yardage, touchdowns, and points. These aggregate totals serve as an excellent baseline, providing the environmental data necessary to refine the model's predictive accuracy. However, even with a perfect situational context, the model remains vulnerable to the physical unpredictability of the players themselves.

As mentioned earlier, another limitation of the model is that it fails to account for injury volatility, specifically by including players who are set to miss the entire season or the inevitable performance drop-off often seen in athletes returning from significant medical setbacks. Fortunately, medical data is well documented in the NFL. This means that players who sustain season-ending injuries before the season begins can be filtered out of the model. Furthermore, specialized injury features can be appended to the dataset, encompassing both recent setbacks and historical career data; this allows the model to learn the correlation between past medical issues and the future durability or performance potential of a player. By integrating these contextual team-level frameworks and injury data, the model can move beyond simple talent identification and begin to account for the complex variables that define professional football.

Conclusion

In this study, I investigated whether algorithmic modeling, specifically through XGBoost regressors, could provide a more objective and accurate alternative to the “hybrid” models used by industry leaders, where algorithmic outputs are “corrected” by human analysts. The results of the study highlight a nuanced performance landscape: while the ESPN consensus remains superior in projecting RB and WR success, the XGBoost regressors were more consistently able to rank QBs and TEs.

This disparity in performance by position likely indicates that the effectiveness of "correction" by a human analyst is heavily dependent on the nature of the data being analyzed. In high-variance positions such as RBs or WRs, the "hybrid" approach is likely benefiting from the ability to understand qualitative changes, such as a shift in philosophical approach on offense or a player’s improved standing in the depth chart, which are not yet reflected in historical data. In contrast, the "pure" approach’s stronger performance in ranking QBs and TEs indicates that in these positions, "correction" is sometimes a source of cognitive interference. Future versions, which incorporate situational context, may further render the results of this study more pronounced.

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